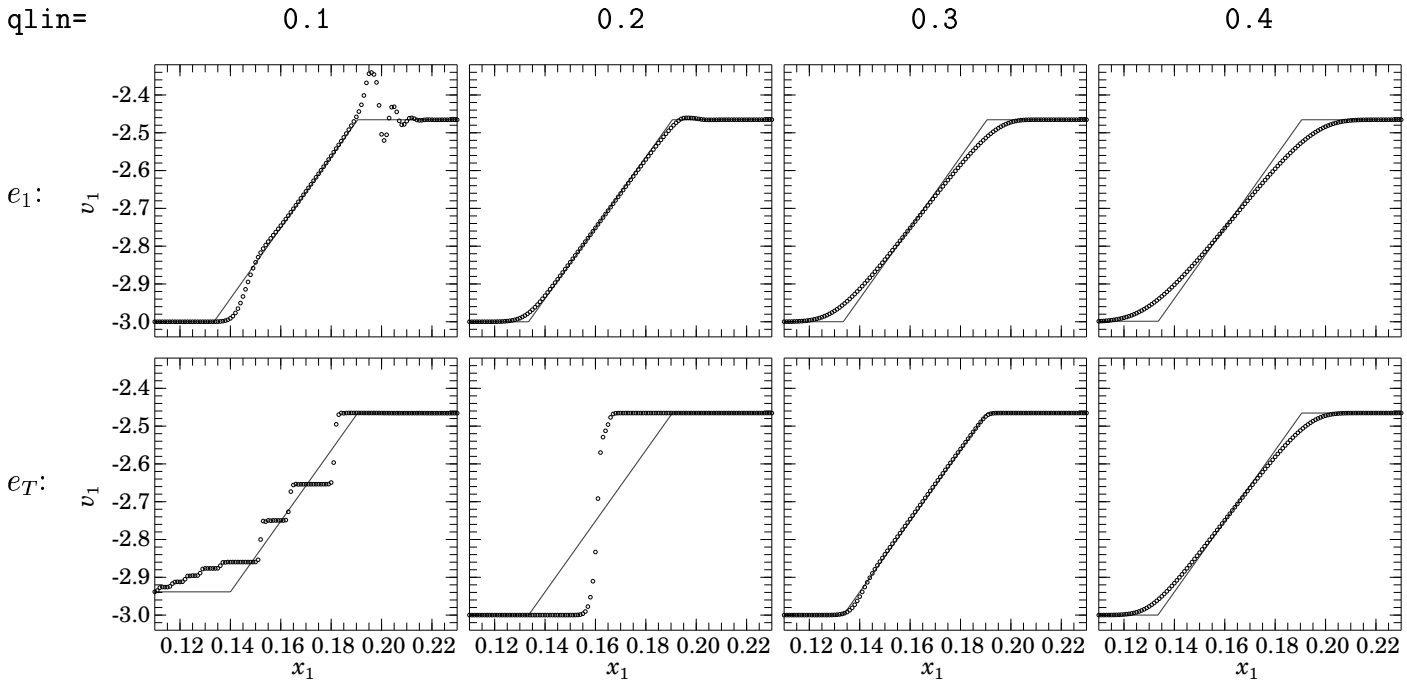
ZEUS-3D 1-D Gallery #13: Rarefaction Shocks



This study is based on Fig. 1 from Falle (2002, ApJ, 577, L123), showing the solution of the HD shock tube test with the left state $(\rho, v_1, v_2, v_3, p_1) = (1, -3, 0, 0, 10)$ and the right state $(0.87469, -2.46537, 0, 0, 8)$ with $\gamma = 5/3$ at time $t = 0.08$. At $t = 0$, the discontinuity is at $x_1 = 0.7$. The grid domain is $0 \le x_1 \le 1$, though only the portion of the grid containing the rarefaction is shown.

Open circles are the `dzeus35` solution for v_1 using 1,000 zones, `CMoC` in its hydrodynamical mode, the internal/total energy equation on the top/bottom row (labelled e_1/e_T), and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.75`, `qcon=0.0`, and `qlin` as indicated at the top of each column. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995, ApJ, 442, 228).

How `dzeus35` captures a rarefaction fan depends on which energy equation is being solved, and the amount of *linear* viscosity (proportional to $\nabla \cdot \vec{v}$; see Stone & Norman, 1992, ApJS, 80, 753) applied, controlled in the code by the internal parameter `qlin`. The traditional von Neumann-Richtmyer artificial viscosity is quadratic in $\nabla \cdot \vec{v}$ and applied only in regions of compression, not expansion such as in a rarefaction. Thus, `qcon`, the `dzeus35` internal parameter controlling the quadratic viscosity, is ineffective in mediating rarefactions and is set to zero in this test.

For sufficient `qlin`, a stable rarefaction is attained regardless of energy equation used. Excessive values of `qlin` just smears the fan beyond its analytical bounds. The internal energy equation can tolerate smaller `qlin` than the total energy equation, and loses stability when `qlin` drops below `0.2` (for this problem). On the other hand, the total energy equation tries to fit "rarefaction shocks" to the rarefaction wave for `qlin < 0.3` (again, for this problem).

Most 1-D test problems in the *ZEUS-3D* 1-D Gallery use `qlin=0.2`, and some even lower without undue effect on the rarefactions (*e.g.*, `qlin=0` in problem #7 in the 1-D Gallery). In large, multidimensional simulations, I often run with `qlin=0` with no evidence of "rarefaction shocks" ever arising in practise. It would seem that the phenomenon of rarefaction shocks may be confined to specialised 1-D shock tube problems and probably of little concern in practise. Still, the next release of this code will remove the last of the operator splitting in the momentum equation which, according to Falle (2002), should eliminate the occurrence of rarefaction shocks even in these test problems.